

TARUNKUMAR BANTUPALLI

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EDUCATION

ARIZONA STATE UNIVERSITY

Masters of Science, Major in Data Science and Analytics

(Data Processing at Scale, Statistical Machine learning, Statistics for Data Analysts, Big Data)

TEMPE, ARIZONA

DECEMBER 2026

CGPA – 3.72/4

SRM UNIVERSITY

Bachelor of Technology, Major in Computer Science; Specialized in Artificial Intelligence

(Python, Deep learning, DSA, Web development, Data Mining, Project budgeting)

INDIA

JUNE 2024

CGPA – 8.82/10

TECHNICAL SKILLS

- **Programming & Scripting:** Python, SQL, PySpark, Bash, Java
- **Data Engineering & ETL:** Airflow, AWS Glue, Lambda, Step Functions, Shell Scripting, Kafka
- **Cloud & Infrastructure:** AWS (S3, Redshift, RDS, EC2, Athena, DMS, EMR), Docker, Terraform
- **Databases:** PostgreSQL, MySQL, MongoDB, Snowflake
- **Analytics & Visualization:** RFM, PCA, DBSCAN, Regression, Tableau, Power BI, Matplotlib, Seaborn

EXPERIENCE

PROMPT ENGINEER

HYDERABAD, INDIA

ZENSAR TECHNOLOGIES PVT LTD – NVIDIA PROJECT

OCTOBER 2024 - NOVEMBER 2024

- Created and verified **400+ high-quality prompts** based on academic book images to support multimodal AI model training, ensuring **100% compliance** with linguistic, contextual, and formatting standards

DATA ENGINEER INTERN

HYDERABAD, INDIA

ELECTRONIC ARTS PVT LTD

FEBRUARY 2024 – OCTOBER 2024

- Built and deployed **ETL pipelines** to ingest **10K+ daily records** from **Asset Panda API** into **MongoDB**, reducing processing time from **2 hours to under 10 seconds**
- Scheduled data pulls using **Task Scheduler** on **VMs**, enabling **100% automated refresh** of inventory data
- Designed **Power BI dashboards** via **ODBC connections** to MongoDB, delivering real-time visibility to **15+ business units** covering **2,000+ employees**
- Streamlined manual updates by developing a **Google Apps Script + JavaScript form** for inventory entry and validating API flows via **Postman**, with detailed logging for traceability
- Developed a **Slack bot** using **Flask**, integrated with internal systems to automate asset tracking, request submissions, and team-level queries
- Enabled self-serve workflows for business unit leads, reducing IT support requests by **60%** and eliminating dependency on email-based communications

PROJECTS

SCALABLE REDDIT ETL PIPELINE ON AWS

- Built a scalable ETL pipeline to process 1M+ Reddit posts/comments, storing raw data in S3 and transforming it via Glue + Athena, reducing processing time by 60%
- Loaded structured data into Redshift, optimizing schema to support sub-second queries across 5+ analytical dimensions (e.g., subreddit, sentiment)
- Automated data workflows with 10+ Airflow DAGs, enabling hourly refresh cycles and fault-tolerant scheduling
- Implemented metadata logging in PostgreSQL for job recovery and validation, ensuring zero data loss

IOT SENSOR DATA PIPELINE FOR SMART ENERGY MONITORING

- Ingested 1M+ IoT energy sensor readings from edge devices via Kafka (or Kinesis) streams into S3 with real-time processing via AWS Lambda
- Aggregated time-series records into hourly/daily summaries using PySpark, and stored in Amazon Redshift
- Built dashboards in Tableau to visualize energy usage trends, anomaly detection, and peak load forecasting
- Achieved <5-minute data latency and enabled real-time alerting on spikes using rule-based triggers

CUSTOMER SEGMENTATION FOR ECOMMERCE MARKETING

- Clustered 541K+ customer records using K-Means, GMM, and DBSCAN to identify marketing personas
- Enhanced segmentation by applying RFM analysis, boosting Silhouette score from 0.38 to 0.44
- Used PCA for cluster visualization and selected optimal models via Elbow method and internal metrics
- Identified high-value cohorts for targeted campaigns, improving cluster cohesion by 30%+

VOLUNTEER EXPERIENCE

STUDENT COUNCIL, SRM UNIVERSITY-AP

SEPTEMBER 2024 – NOVEMBER 2023

- Developed a responsive, mobile-friendly event website using HTML5, CSS3, JavaScript, PHP, and Bootstrap, attracting 10,000+ visits and 7,200+ registrations in the first week